Universität Zürich[UZH] | Institute of Education

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The challenge of studying a technology's learning benefits

Media comparison studies of GenAI in education

Prof. Dr. Joshua Weidlich, University of Zurich | Zurich University of Teacher Education

Open research seminar · AI Lab for Research on Teaching and Learning · Tallinn University

What is the impact of AI on learning?

New technology, old questions

Film	Radio	Television	Computer	Internet	VR	GenAI
1920s	1930s	1950s	1980s	2000s	2010s	2020s

“Is the new better than the old? Whoooo!!”



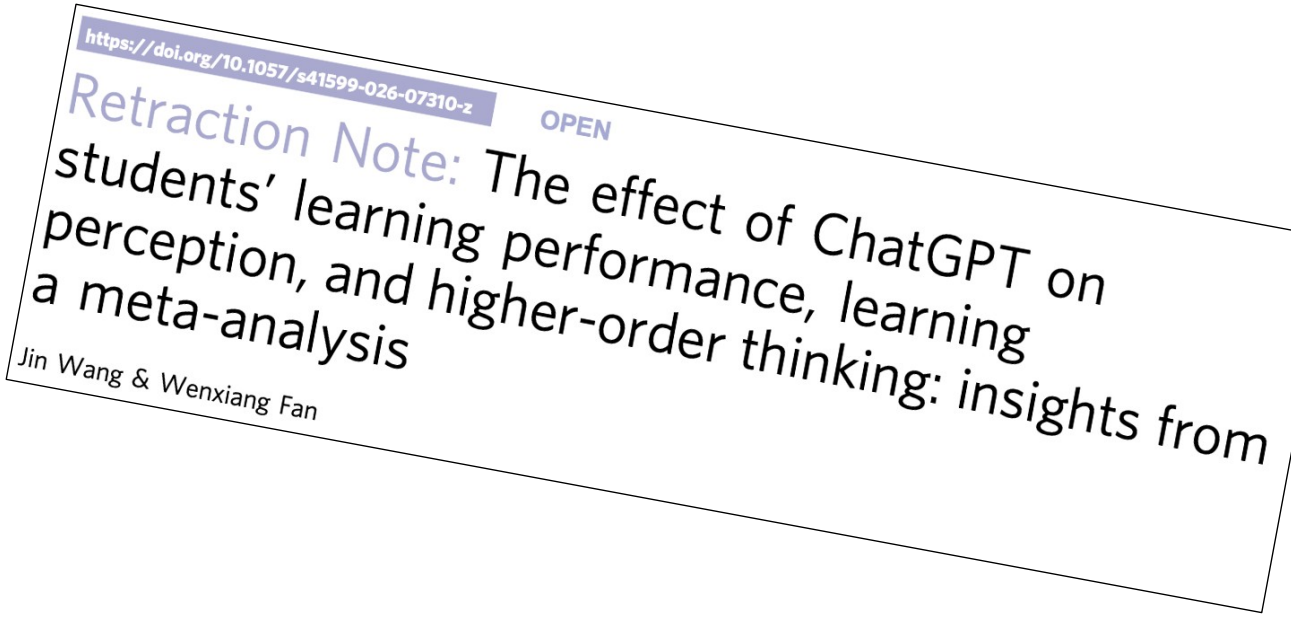
A widely cited answer

g = 0.87

ChatGPT → students' learning performance

Wang & Fan (2025) · 51 studies · 450+ citations

RETRACTED · 22 APRIL 2026



Humanities and Social Sciences Communications, 12, 621. Retraction note: doi.org/10.1057/s41599-026-07310-z

Richard Clark (1983)

"...media are mere vehicles that deliver instruction but do not influence student achievement any more than the truck that delivers our groceries" (p. 445)

Richard E. Clark (1983), *Review of Educational Research, 53*(4)



Positive findings, then, are due to *confounding*.

What is *confounding*?

— Confounding: a third variable affects both IV and DV

— Result: a “spurious association”

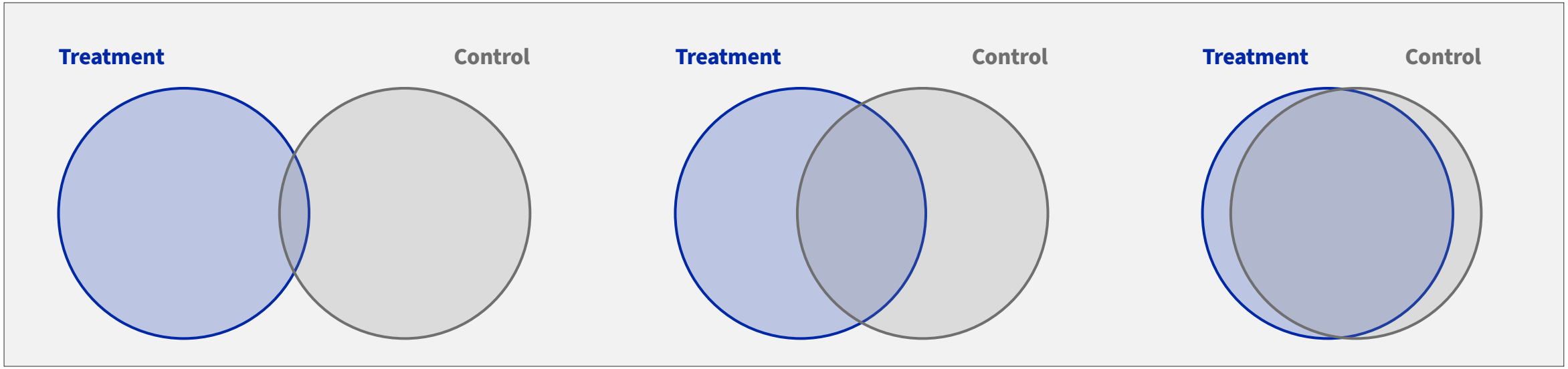
— Example: *negative* association between study time and test scores

What is *confounding*?

— Confounding: a third variable affects both IV and DV

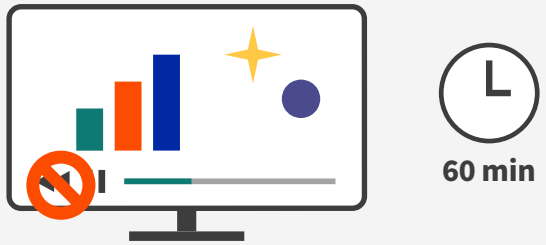
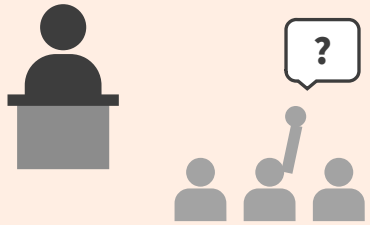
— Result: a “spurious association”

— Example: negative association between *study time* and *test outcome*



An example: explainer video vs. lecture

“Do students learn better from an explainer video than from a lecture?”

	Naive comparison	Learning time controlled	(Almost) everything controlled
VIDEO			
LECTURE			
	Differences – but why?	**A bit cleaner …**	**Clean – but what’s left?**

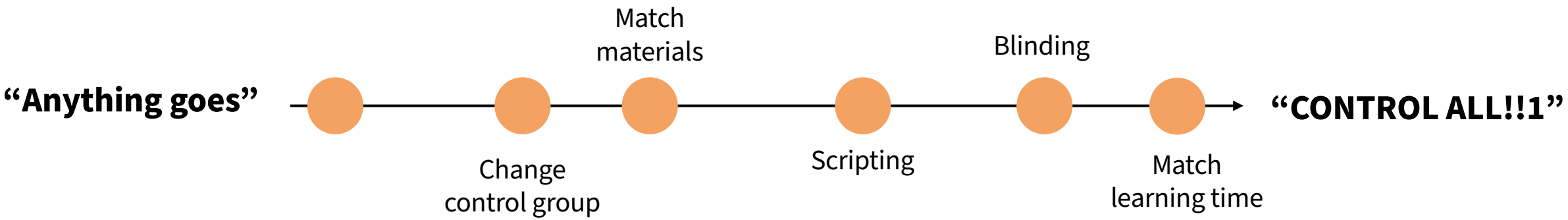
The dilemma of media comparisons

LITTLE CONTROL

Differences – but why?

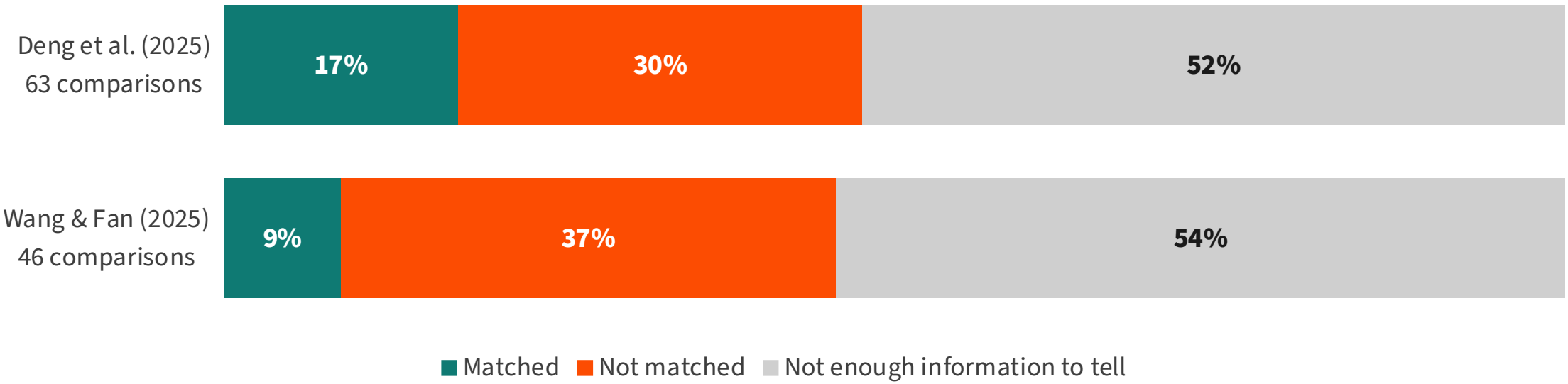
STRONG CONTROL

Cleaner – but what's left?



Most comparisons cannot isolate the medium

Were instructional methods matched between the ChatGPT and the control condition?



Where methods differed, they almost always favoured the ChatGPT condition (18 of 19; 16 of 17)

Lawson, Martella, Weidlich, Mulders & Buchner (2026), Computers & Education, 254, 105701

Auditing the meta-analyses

0 of 14

meta-analyses on AI in education provided a valid basis for their claims

together cited 2,000+ times

— No coherent construct

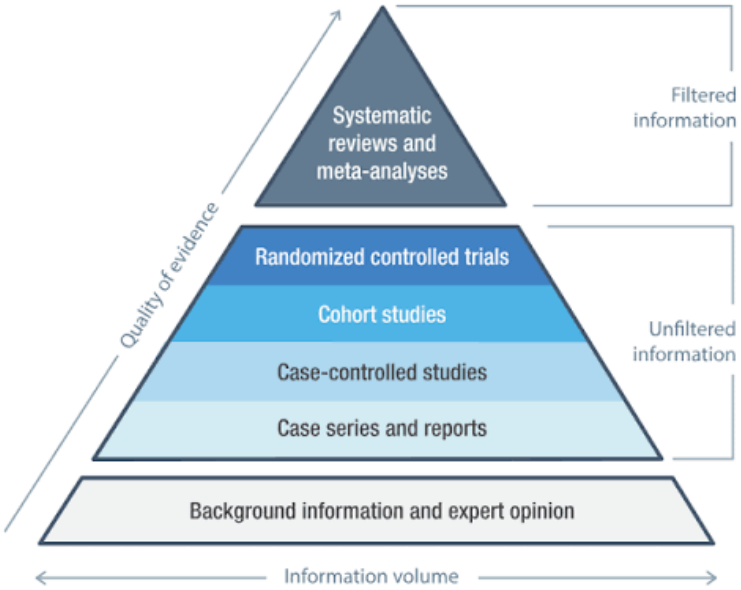
— Publication bias insufficiently assessed

— Severe heterogeneity throughout

— 61% of vetted primary studies problematic

O’Neill (2026). Presumed effective: The manufacturing of an evidence base for AI-in-education through flawed meta-analysis [Preprint]

Rigor Laundering: **Meta-analyses inherit the weaknesses of the primary literature (GIGO) – while projecting methodological robustness.**



Four questions for interpreting media comparisons

1. **What exactly was the intervention?**
2. **Compared to what?**
3. **What was learned, and how was it measured?**
4. **Which claim does it support?**

Building on Weidlich, Gašević, Drachsler & Kirschner (2025), Journal of Computer Assisted Learning

What exactly was the intervention?

“Students used ChatGPT for 30 minutes”



TOOL

model, version, configuration

×

TASK

what learners were asked to do

×

METHOD

scaffolds, prompts, instructions

×

ENACTMENT

what learners actually did

Compared to what?

Compared to …	… the study answers the question
regular teaching (“business as usual”)	Is it better than what would have happened otherwise?
the best available alternative	Is it better than the best we already have?
the same method without AI	Does it matter that AI delivers the method?
a standard chatbot	Does design X add anything beyond “regular” AI use?
the same AI without feature X	What does feature X contribute?

cf. Freedland et al. (2019) on choosing comparison conditions

What is “business as usual”, anyway?

Medicine

Standard of Care

Treatment vs. best proven treatment

Placebo or “nothing” only if no proven treatment exists.

Declaration of Helsinki (WMA)

Education

“Traditional instruction”

Often a blank: the control condition is barely described

Heterogeneous, for good didactic reasons

Lawson et al. (2026)

cf. Freedland et al. (2019) on “usual care” as a comparison condition

What was learned, and how was it measured?

	IMMEDIATELY	AFTER A DELAY
WITH THE TOOL	**Short-term performance** many GenAI studies sit here	**Sustained performance**
WITHOUT THE TOOL	**Learning**	**Durable learning & transfer**

Which claim does it support?

~~“AI improves learning”~~

This [package] improved [outcome] for [learners] compared to [alternative] over [time span].

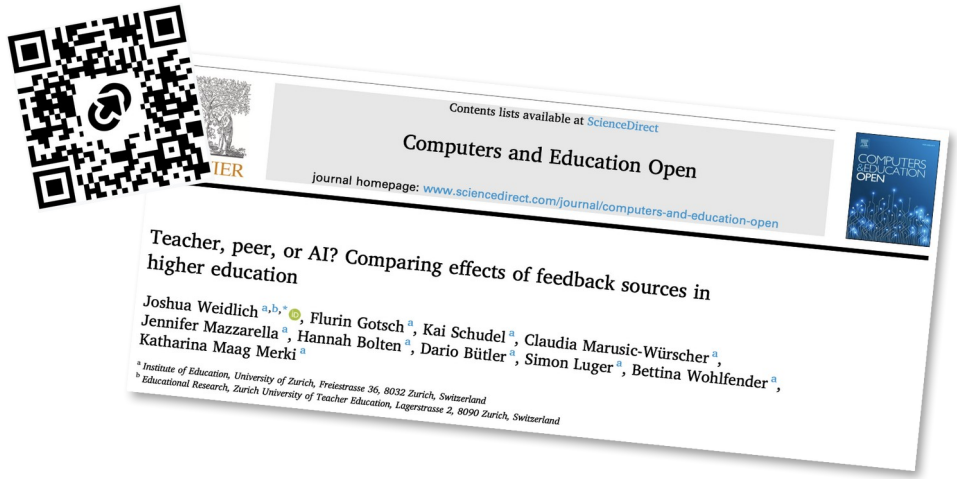
Which claim needs which design?

Claim	Design that can support it
Programme effect	Pragmatic RCT or quasi-experiment; realistic comparison; analysed as assigned
Design principle	AI vs. AI: A/B tests, component or factorial designs
Mechanism	Process data; theory-based mediators and moderators
Description	Descriptive, interpretive, design-based research
~~**Medium effect**~~	– reformulate into one of the above

Every study answers a question.

But which one?

Example 1: Feedback sources



Initial submission → Everyone writes peer feedback → **Receive feedback Teacher/Peer/LLM** → Questionnaire (FPQ, …) → Revise & resubmit





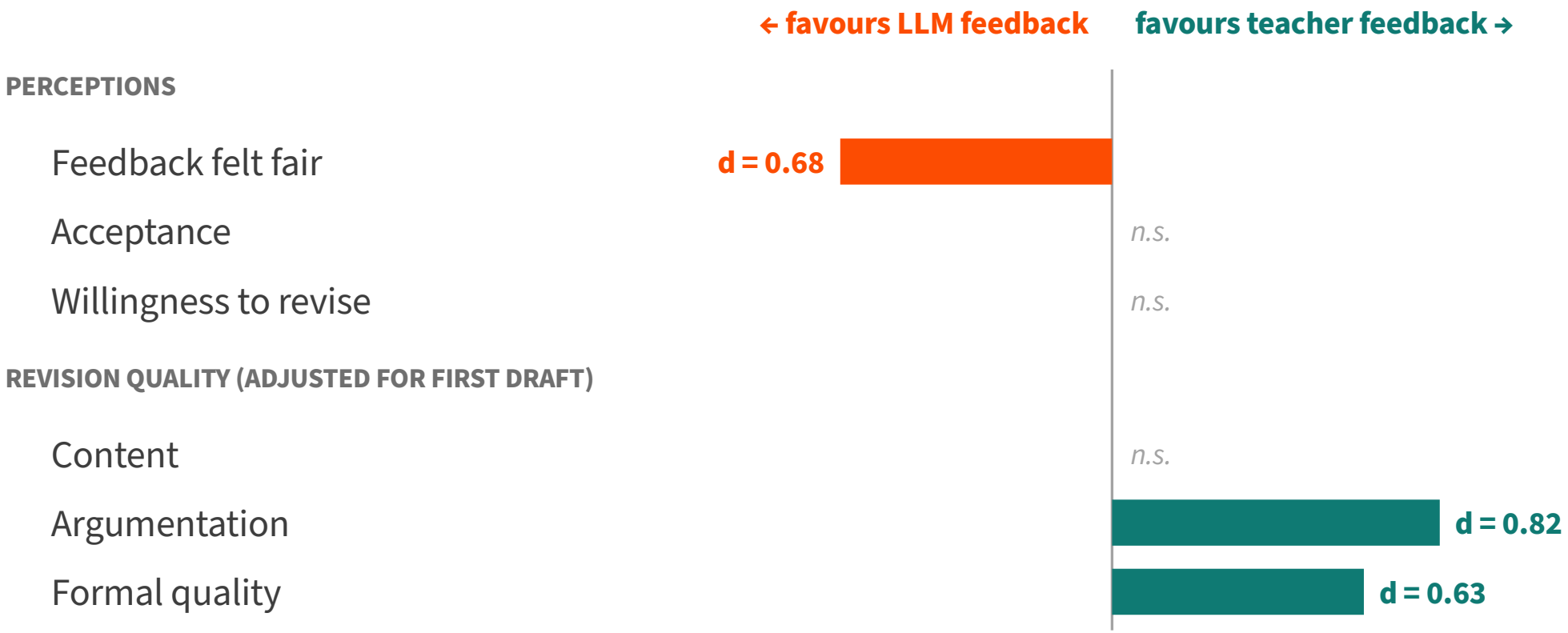
— N = 90 bachelor students

— authentic written assignment on school quality development

— identical rubric template & feedback instructions

Perceptions and revision diverge

Teacher feedback compared with LLM feedback (effect sizes, Cohen's d)

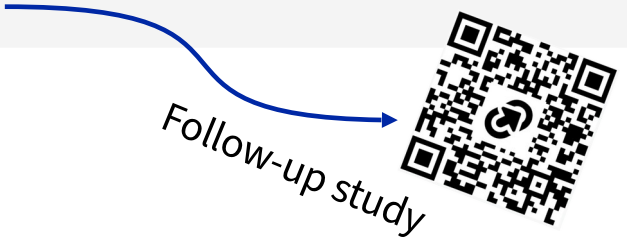


Peer feedback: accepted best, and as effective as teacher feedback for formal quality

Weidlich et al. (2025). Teacher, peer, or AI? Comparing effects of feedback sources in higher education. Computers and Education Open, 9, 100300 · N = 90

Example 1 through the four questions

1	**What exactly was the intervention?**	Not really “the source”, but a text with a bundle of contents and features
2	**Compared to what?**	Realistic alternatives: teacher and peer feedback – same rubric, source blinded
3	**What was learned, and how was it measured?**	Perceptions and immediate revision quality – not learning
4	**Which claim does it support?**	Feedback effects plausibly identified. But still a package: it’s unclear what in the feedback messages made the difference.



Example 2: AI tutor vs. in-class active learning

AI TUTOR (AT HOME)

Pre-recorded introduction video

Custom-built GPT-4 tutor

Expert system prompt + step-by-step solutions

Self-paced

ACTIVE LEARNING (CLASSROOM)

Live introduction by the instructor

Peer instruction in small groups

Support from course staff + instructor feedback

Fixed class time

Same worksheet, content and learning goals

N = 194, introductory physics (Harvard) – randomized crossover design over two lessons (pre/post)

Kestin, Miller, Klales, Milbourne & Ponti (2025), Scientific Reports, 15, 17458

“AI tutoring outperforms in-class active learning”

>2×

median learning gain

post-test median 4.5 vs. 3.5 (pre-test 2.75)

49 min

median time with the tutor

time on task unrelated to post-test score

4.1 vs. 3.6

self-reported engagement

1–5 scale; motivation also higher

Kestin et al. (2025). AI tutoring outperforms in-class active learning […]. Scientific Reports, 15, 17458

What exactly was compared?

One lesson per condition, same worksheet – what else differed?

	Content	Introduction	Guidance	Working	Setting	Time
AI TUTOR	Same worksheet and learning goals	Pre-recorded video	GPT-4 tutor with pre-written solutions	Alone	At home	Self-paced, median 49 min
ACTIVE LEARNING		Live, by the instructor	Course staff, then instructor feedback	In groups of 2–3	In the classroom	Fixed, about 60 min
	=	≠	≠	≠	≠	≠

The AI is one of five differences. Which one did the work?

Kestin, Miller, Klales, Milbourne & Ponti (2025), Scientific Reports, 15, 17458 – Methods

Example 2 through the four questions

1	**What exactly was the intervention?**	A bundle: LLM + expert prompt + pre-written solutions + video + self-pacing at home
2	**Compared to what?**	A strong comparison with the same worksheet – but the AI was one of five differences
3	**What was learned, and how was it measured?**	Immediate post-test in the same session; ceiling effects; no retention measure
4	**Which claim does it support?**	This specific AI tutoring package beat another strong form of instruction on an immediate test. (Which element did the work?)

Study the principle,

not a technology or a product.

Discussion

— When people ask about the effects of AI, what is the unspoken reference – compared to what?

— What must – and must not – be controlled for with AI? What are its essential affordances?

— Should we as researchers answer naïve media-effect questions – or is it our job to reframe them productively?

— Are media comparisons beyond rescue?

Slides, handout & reading list
joshuaweidlich.github.io/updates/tallinn-talk.html

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Thank you

joshua.weidlich@ife.uzh.ch

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Feedback 1

Feedback 2

Confounder

ChatGPT

Clark’s replaceability challenge

“If a treatment can be replaced by another treatment with similar results, the cause of the results is in some shared (and uncontrolled) properties of both treatments.”
Clark (1994, p. 22)



— Could another medium, or a person, deliver the same method? Then the result is about the method.

— If AI does something that cannot be replaced: name that attribute and test it directly.

Clark, R. E. (1994). Media will never influence learning. Educational Technology Research and Development, 42(2), 21–29.

Value-added designs

Compare the same tool with and without one feature (Mayer, 2019)

Media comparison	**AI tutor**	vs.	**Regular lesson**	**many differences at once**
Value-added	**AI tutor**	vs.	**AI tutor + one feature** e.g. hints instead of answers	**one difference**

— Supports a design-principle claim: “feature X helps, with this tool, task and group”

— Example: GPT Base vs. GPT Tutor – same model, different prompting (Bastani et al., 2025)

— Fits inside a rollout: randomise one feature, test without AI, keep the logs

Mayer, R. E. (2019). Computer games in education. Annual Review of Psychology, 70, 531–549 · Bastani, H., et al. (2025). PNAS, 122(26).

Was the activity comparable?

What students do cognitively in each condition (ICAP; Chi & Wylie, 2014)



— STEM meta-analysis: 33 studies showed large effects only because the activities in the two groups were not comparable

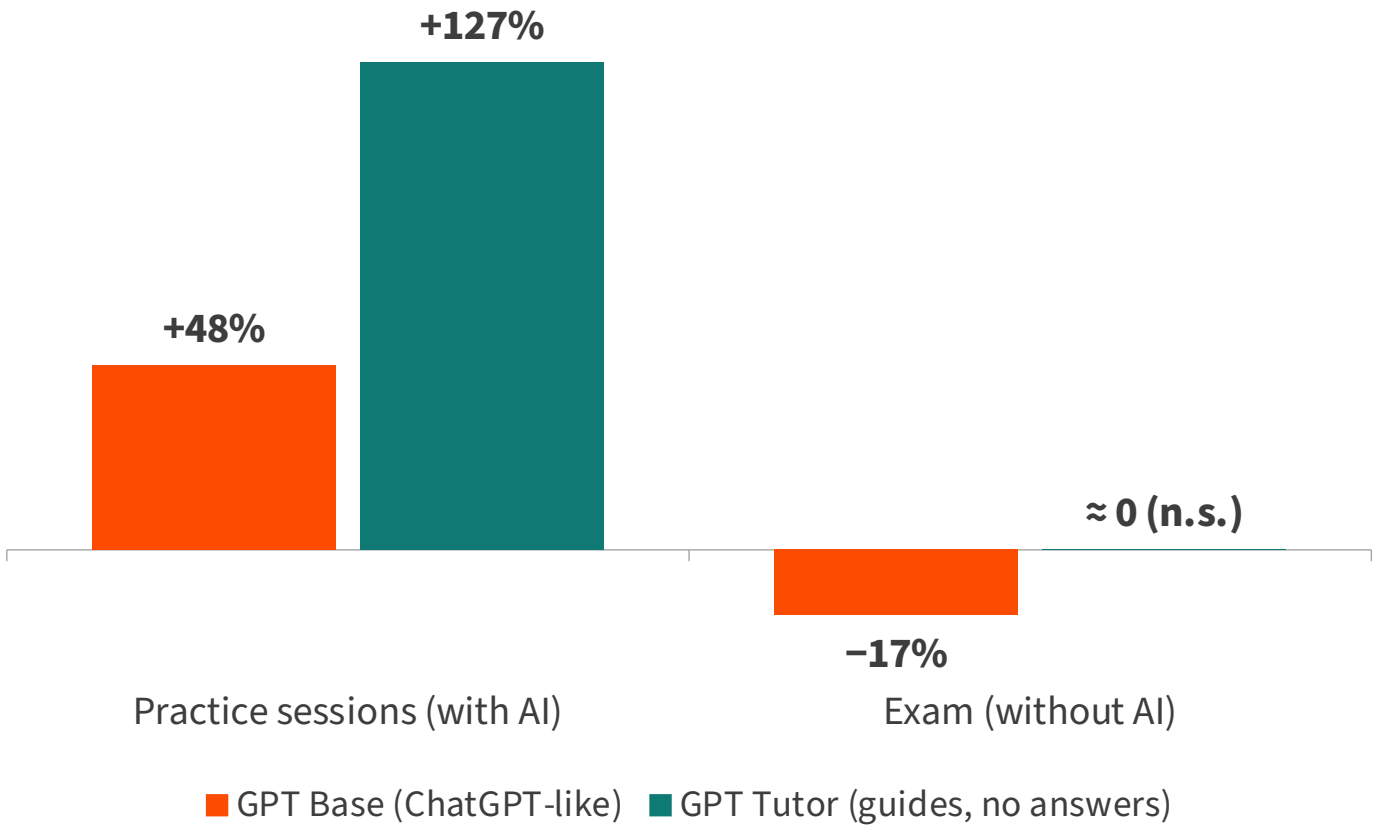
— GenAI looked promising where it added to students' own activity; replacing that activity can be detrimental

— Ask: what were students doing, minute by minute, in each condition?

Küchemann, S., et al. (2026). Artificial Intelligence Review · Chi, M. T. H., & Wylie, R. (2014). Educational Psychologist, 49(4), 219–243.

Performance with AI, learning without it

Change compared with the no-AI control group



— ~1,000 high school maths students in Turkey

— GPT-4 available during practice sessions

— Exam taken without AI

— GPT Base students felt they had learned as much as the others

Bastani, H., et al. (2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. PNAS, 122(26).

Expectations in active controls

AI group vs. an active control group doing something else for the same time

Time on task	✓ **matched**
Activity and attention	✓ **matched**
Expected benefit	× **usually not matched**

— Participants know which condition they are in, and expectations of AI are high

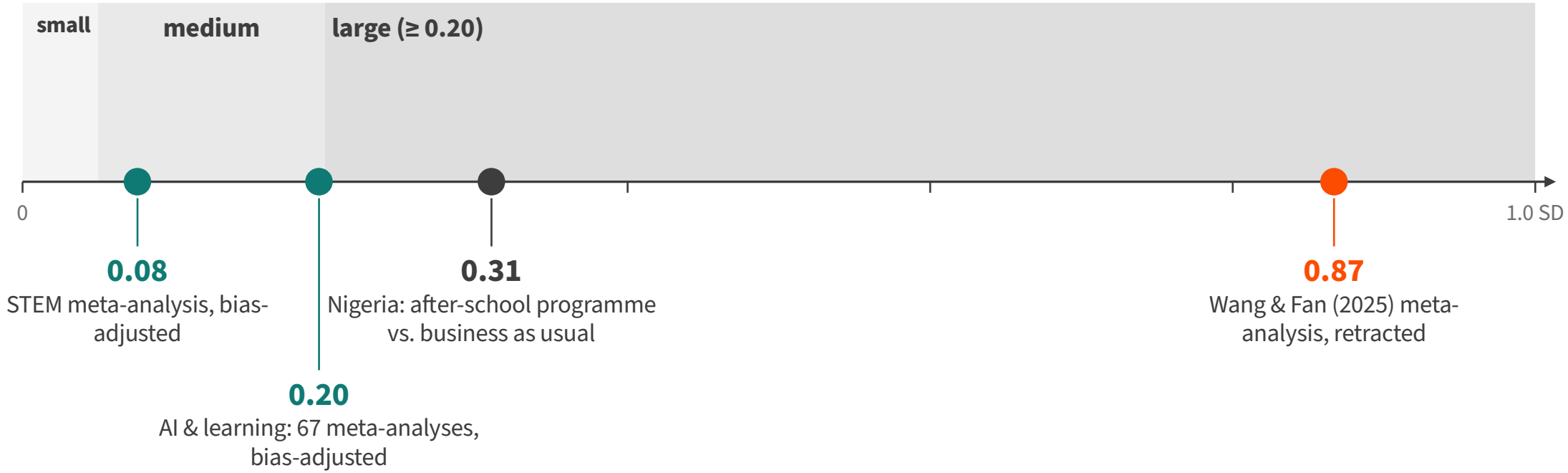
— Only when both groups expect the same improvement can a difference be attributed to the treatment

— Options: measure expected benefit in both groups; compare AI variants, where expectations differ less

Boot, W. R., Simons, D. J., Stothart, C., & Stutts, C. (2013). The pervasive problem with placebos in psychology. Perspectives on Psychological Science, 8(4), 445–454.

How big is a big effect?

Benchmarks for causal education studies with standardised achievement tests (Kraft, 2020)

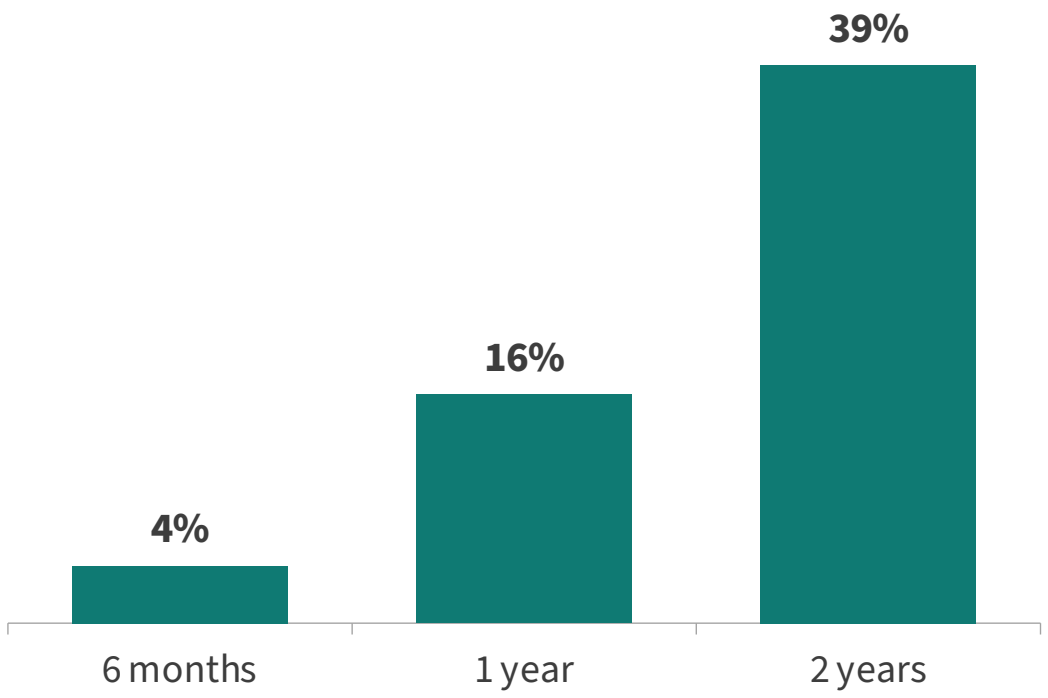


— Small samples, researcher-made tests and developer-run evaluations tend to show larger effects

Kraft (2020), Educational Researcher, 49(4) · Cheung & Slavin (2016), Educational Researcher · Wolf et al. (2020), JREE · Küchemann et al. (2026), Artificial Intelligence Review · Bartoš et al. (2026), preprint · De Simone et al. (2025), World Bank WP 11125 · Wang & Fan (2025), retracted 2026. Different outcomes and designs: for orientation only.

The shelf life of LLM findings

Share of claims about LLMs that no longer held, by time since publication



— 50 widely cited claims, re-tested on 20 models released July 2024 to September 2026

— Overall: 44% expired, 34% weakened, 16% persist, 6% strengthened

— Example: 55% fabricated citations with GPT-3.5 (2023), about 1% or less with current models

— For education studies: report model, version, date, prompts, interface and access rules

zou-group.github.io/llm-claims-shelf-life

Queen, O., & Zou, J. (2026). The shelf life of LLM findings. Claims about LLM behaviour in general, not learning outcomes; 50 claims, wide confidence intervals.

What a programme evaluation can show

GPT-4 after-school tutoring in Nigeria, randomised (De Simone et al., 2025)

Supported

— The programme beat business as usual: 12 teacher-guided 90-minute sessions with GPT-4

— +0.31 SD on the end-of-programme test; +0.23 SD in English

Not supported

— That AI did the work: extra time, teacher guidance and structured tasks came with it

— The dose–response (+0.031 SD per day attended): attendance was not randomised

— Randomised among the 52% of eligible students who signed up

De Simone, M., et al. (2025). From chalkboards to chatbots: Evaluating the impact of generative AI on learning outcomes in Nigeria. World Bank Policy Research Working Paper 11125.